

		parallel to the lamps P and Q does nothing to change the circuit itself
22	D	Total energy provided by battery is 60J Emf is $W/Q = 60/20 = 3.00 \text{ V}$ Pd across $r = 10/20 = 0.50 \text{ V}$ Pd across $R = 50/20 = 2.50 \text{ V}$
23	A	The forces acting on XY and ZW have the same magnitude in opposite